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MLAIRE: Multilingual Language-Aware Information Retrieval Evaluation Protocol

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Abstract

Multilingual Information Retrieval is increasingly important in real-world search settings, where users issue queries over mixed-language corpora. Existing evaluations mainly reward language-agnostic semantic relevance, treating relevant passages equally regardless of language. Yet retrieval utility also depends on the language of the retrieved passages: users may prefer results they can read and verify in the query language, and query–passage language mismatch can complicate downstream grounding and answer verification in Retrieval-Augmented Generation systems. To evaluate this language-aware dimension, we introduce MLAIRE, a **M**ultilingual **L**anguage-**A**ware **I**nformation **R**etrieval **E**valuation protocol that disentangles cross-lingual semantic retrieval from query-language preference. MLAIRE constructs controlled pools with parallel passages across languages, enabling measurement of semantic retrieval accuracy and query-language preference when equivalent translations are available. We propose language-aware metrics, including Language Preference Rate (LPR) and Lang-nDCG, together with a 4-way decomposition separating semantic and query-language preference failures. Evaluating 31 dense, sparse, and late-interaction retrievers, we show that standard metrics obscure distinct behaviors: semantically strong retrievers may return correct content in a non-query language, while retrievers with stronger query-language preference may retrieve less semantically relevant passages.

1 Introduction

Multilingual Information Retrieval (MLIR) aims to retrieve relevant content when queries and documents appear in diverse languages [1, 2]. This setting is central to real-world search and Retrieval-Augmented Generation (RAG) systems, where users issue queries over corpora containing content in many languages. Accordingly, MLIR systems are primarily evaluated by whether they can identify semantically relevant content across languages [3–5].

However, semantic retrieval quality alone does not determine whether a retrieved result is useful to the user. When several language versions of the same relevant content are available, a retriever may rank non-query-language content above its query-language counterpart. This creates a gap between semantic retrieval and query-language preference: the former asks whether the retrieved content is relevant, while the latter asks whether the relevant content is written in the query language.

This gap is visible even among strong multilingual retrievers. The scatter plot in Figure 1 compares standard nDCG with Language Preference Rate (LPR), which measures whether the query-language version of the target content is scored above its semantically equivalent alternatives. The results are macro-averaged over the three MLAIRE datasets. In this figure, PPLX-Embed-4B and BGE-M3 achieve strong standard retrieval performance but show lower LPR than mE5-large, while BM25

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and OpenSearch-NSE exhibit high LPR despite weaker nDCG. This suggests that standard retrieval metrics can obscure whether a retriever prioritizes query-language evidence.

This distinction matters in practical multilingual retrieval scenarios. The language of the retrieved passage affects whether users can read, verify, and act on the result. It can also affect downstream RAG behavior: when the query and retrieved passages are written in different languages, the generator must interpret cross-lingual evidence while producing an answer in the user's language. Recent studies on multilingual and cross-lingual RAG report that query–context language mismatch can degrade answer correctness and make models less likely to preserve the expected response language [6–8]. Thus, query-language preference in retrieval is not merely a user-facing preference, but also a retrieval behavior that can affect downstream answer generation.

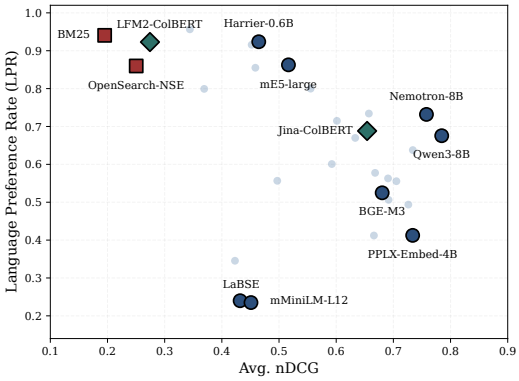

Figure 1: Each point shows a retriever evaluated on MLAIRE, with average nDCG on the x-axis and Language Preference Rate (LPR) on the y-axis.

To evaluate this behavior, we introduce **MLAIRE**, a Multilingual Language-Aware Information Retrieval Evaluation protocol. Inspired by the parallel-corpus formulation of Roy et al. [9], MLAIRE constructs candidate pools where semantically equivalent passages coexist across languages. This setup preserves standard MLIR evaluation of cross-lingual semantic retrieval while making query-language preference directly observable. Using MLAIRE, we report conventional retrieval metrics together with language-aware diagnostics, including LPR, Lang-nDCG, and a 4-way decomposition of rank-1 outcomes. Our analysis shows that query-language preference constitutes a distinct and structured dimension of MLIR behavior, rather than a property captured by language-agnostic retrieval metrics alone.

Our main contributions are as follows:

- We formulate query-language preference as an important aspect of MLIR evaluation.
- We introduce **MLAIRE**, a language-aware evaluation protocol with metrics and diagnostics for analyzing retrieval behavior beyond semantic relevance.
- We evaluate 31 retrievers across dense, sparse, and late-interaction architectures, revealing systematic mismatches between semantic retrieval quality and query-language preference.

2 Background

2.1 Multilingual Information Retrieval (MLIR)

Definition of MLIR Information Retrieval (IR) settings can be distinguished by the language relationship between the query and the candidate corpus [10, 11]. Monolingual IR assumes that queries and relevant documents are written in the same language [12, 13], while Cross-Lingual IR (CLIR) considers settings where the query and target documents are written in different languages [14, 15]. In this paper, we adopt the definition of MLIR following the Cross-Language Evaluation Forum (CLEF): a task in which queries are issued in different languages and the candidate pool may contain relevant passages in multiple languages simultaneously [10]. Unlike Multi-Monolingual IR, where each query is evaluated against a same-language corpus [16], MLIR allows passages in multiple languages to coexist within a shared candidate pool.

Conventional Evaluation of MLIR Existing MLIR evaluations primarily adopt a language-agnostic view of relevance. A retrieved passage is considered relevant if it satisfies the information need, regardless of the language in which it is written [9, 11, 17]. This view is essential for evaluating cross-lingual semantic retrieval: a retriever should be able to recognize that semantically equivalent texts in different languages express the same information. This objective is consistent with multilingual representation learning, which aims to place semantically equivalent texts from different languages in a shared embedding space [18, 19].

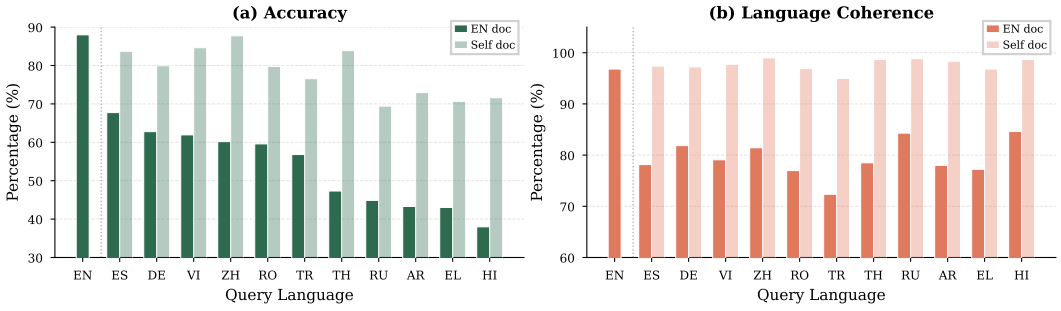

Figure 2: RAG experiment on XQuAD across 12 query languages. We evaluate Qwen2.5-7B-Instruct using Exact Match (EM) and response-language correctness.

However, this language-agnostic view does not distinguish between different language versions of the same relevant content. When the query-language passage and its non-query-language translations are all semantically relevant, standard metrics such as Recall and nDCG treat them as equally relevant [20, 21]. As a result, existing evaluations can determine whether a retriever finds the right content, but not whether it prioritizes the version written in the query language. We therefore evaluate query-language preference as a complementary axis alongside language-agnostic semantic retrieval.

2.2 Query-Language Preference as an Evaluation Dimension

Query-language preference matters because the language of a retrieved passage is part of retrieval utility. A passage can be semantically relevant but still difficult for the user to read, verify, or act on if it is written in a language the user did not use or cannot readily understand. This issue is especially important in user-facing search, where prior studies on multilingual production search systems show that users prefer results written in the language of their query [22–26]. In this sense, query-language preference reflects a legitimate aspect of user intent.

The same issue also arises in Retrieval-Augmented Generation (RAG) systems, where retrieved passages are directly consumed by a generator. If the query and retrieved passage are written in different languages, the generator must interpret cross-lingual evidence while producing an answer in the user's language. Recent studies on multilingual and cross-lingual RAG report that query–context language mismatch can degrade answer correctness and make models less likely to preserve the expected response language [6–8].

To further examine this effect, we conduct a controlled RAG experiment using XQuAD [27] and Qwen2.5-7B-Instruct [28]. For each query language, we compare two conditions with gold passages of the same meaning: an English passage and a passage written in the query language. Figure 2 reports answer accuracy in (a) and language coherence in (b), where language coherence indicates whether the model answers in the query language. When the query is in English, providing the English relevant passage yields high accuracy and high language coherence. However, when non-English queries are paired with English relevant passages of the same meaning, both answer accuracy and language coherence drop substantially. Replacing the English passage with the query-language passage consistently improves performance and makes the model much more likely to answer in the query language. Since the two conditions provide equivalent gold content, this gap shows that the language of the retrieved passage matters even when semantic relevance is controlled.

These observations motivate a language-aware evaluation perspective that measures whether retrievers prioritize query-language evidence when equivalent relevant passages are available across languages.

3 MLAIRE

MLAIRE follows a simple design principle: each query is paired with a candidate pool that contains semantically equivalent relevant passages in multiple target languages. This construction creates a controlled retrieval setting where a model's ranking behavior reveals two complementary properties. The first is its ability to retrieve semantically relevant content across languages, and the second is its tendency to prioritize query-language evidence when multiple relevant translations are available.

Table 1: Summary of the evaluation datasets used in **MLAIRE**

Dataset	# Languages	# Queries	# Passages	Max N_{rel}	Parallelism
Belebele	122	109,800	59,536	122	Full
XQuAD	12	14,280	2,880	12	Full
MLQA	7	42,245	35,145	4	Partial

3.1 Evaluation Dataset Construction

We build the evaluation dataset from three multilingual resources containing (partially) parallel passages. For each dataset, we organize passages into content groups, where each group contains passages expressing the same underlying content in different languages. For a query q, all passages in its target content group are treated as semantically relevant, regardless of language. Among them, the passage written in the same language as q is treated as the query-language relevant passage.

Belebele [29] provides 488 reading-comprehension passages derived from FLORES-200 [30], each translated into 122 language variants, with associated queries. **XQuAD** [27] provides 240 parallel paragraphs and 1,190 extractive QA pairs in each of 12 languages. **MLQA** [31] provides partially parallel extractive QA data across 7 languages; unlike Belebele and XQuAD, its passages are not fully parallel, and most groups contain translations in only 3–4 languages rather than all 7. We note that Belebele, XQuAD, MLQA are all utilized in MMTEB [16]. Table 1 summarizes the evaluation dataset construction of MLAIRE.

3.2 Language-Aware Metrics

We report standard retrieval metric (nDCG@k) and language-aware metrics designed to capture query-language preference behavior. For a query q, let ℓ_q denote its language and g_q denote its target content group. Each content group consists of semantically equivalent passages across languages. For a passage d, let ℓ_d and g_d denote its language and content group, respectively.

Language Preference Rate (LPR) Let Q denote the set of evaluation queries. For each query $q \in Q$, let d_q^* denote the highest-scoring passage among passages in the target content group:

$$d_q^* = \arg\max_{d: g_d = g_q} s(q, d),$$

where $s(q, d)$ is the retriever score. We define LPR as

$$\text{LPR} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\left[\ell_{d_q^*} = \ell_q\right]$$

LPR measures how often the query-language version of the target content is scored above its cross-lingual alternatives.

Lang-nDCG@k To evaluate language-aware ranking quality, we assign higher relevance to passages that both match the target content group and are written in the query language:

$$\text{rel}_{\text{lang}}(d, q) = \begin{cases} 3, & \text{if } g_d = g_q \text{ and } \ell_d = \ell_q, \\ 2, & \text{if } g_d = g_q \text{ and } \ell_d \neq \ell_q, \\ 0, & \text{otherwise} \end{cases} \qquad \text{DCG@}k = \sum_{i=1}^{k} \frac{2^{\text{rel}_{\text{lang}}(d_i, q)} - 1}{\log_2(i+1)}$$

We compute Lang-nDCG@k by normalizing DCG@k with the maximum possible DCG (IDCG) under this language-aware grading scheme. Unlike standard nDCG@k, Lang-nDCG@k distinguishes the query-language version of the target content from its cross-lingual alternatives.

3.3 Top-1 4-way Decomposition

To diagnose retrieval behavior, we classify the top-1 ranked passage by semantic relevance and query-language match. This yields four mutually exclusive outcomes: (1) `perfect`, if the passage

is semantically relevant and in the query language; (2) `lang_fail`, if it is semantically relevant but in a different language; (3) `sem_fail`, if it is in the query language but semantically irrelevant; and (4) `both_fail`, if it is neither semantically relevant nor in the query language. Aggregating these outcomes across queries reveals whether rank-1 errors are primarily driven by language mismatch, semantic mismatch, or both. This analysis complements LPR: while LPR compares language preference among semantically equivalent target passages, the rank-1 decomposition characterizes the actual first result returned from the full candidate pool.

4 Experimental Setup

4.1 Retrieval Paradigms

We evaluate 31 publicly available multilingual retrievers across three retrieval paradigms: dense, sparse, and late-interaction. This pool covers a broad suite of multilingual embedding models ranging from 100M to 8B parameters, alongside two sparse baselines and two late-interaction retrievers. Our selection prioritizes breadth across paradigms and model scales, and includes widely used or recently released publicly available multilingual retrievers at the time of our experiments.

Dense Our dense retrievers encompass diverse model lineages, ranging from widely adopted encoder-only families such as multilingual-e5 [32], bge-m3 [33], gte [34], snowflake-arctic [35], nomic-embed [36], embeddinggemma [37] and jina [38, 39], to recent LLM-based embedding models including Qwen3-Embedding [40], llama-nemotron [41], and pplx-embed [42]. In this paradigm, queries and passages are independently encoded into fixed-dimensional vectors and scored by cosine similarity. We use each model's prescribed pooling strategy (CLS, mean, or last-token) and follow their recommended instruction templates; representative prefix formats are listed in Appendix C.

Sparse We evaluate two multilingual sparse retrievers: a subword lexical baseline and a neural sparse model. For BM25, we tokenize queries and passages with the XLM-RoBERTa tokenizer [43] and index with Lucene-style BM25 (k_1=1.2, b=0.75) [44]. For the opensearch-neural-sparse-encoding-multilingual-v1 [45], queries and passages are encoded into learned sparse vectors over the MLM vocabulary, and scored by inner product.

Late-Interaction For late-interaction retrieval, we evaluate jina-colbert-v2 [46] and LFM2-ColBERT-350M [47], both of which build upon the late-interaction architecture [48]. Queries and passages are represented as token-level vectors and scored with MaxSim, which sums the maximum similarity between each query token and all passage tokens. For scalable and efficient evaluation, passages are indexed using the PLAID engine [49].

4.2 Evaluation Protocol

We evaluate each retriever on the three datasets of MLAIRE independently: for every (model, dataset) pair, we encode the dataset's full corpus and retrieve the top-k passages per query. The retrieval depth k is chosen per dataset so that it exceeds the maximum number of relevant passages per query: we use k=20 for MLQA (at most 4 relevant passages per query) and XQuAD (12 relevant passages per query), and k=200 for Belebele, whose 122-language parallel structure admits up to 122 relevant passages per query. Because LPR compares the relative scores of semantically equivalent target passages, we compute LPR using scores for all passages in the target content group, independent of whether those passages appear in the retrieved top-k list.

Full hardware configuration, dependency partitioning across virtual environments, and reproducibility artifacts are described in Appendix D.

5 Results and Analysis

We evaluate 31 retrievers on **MLAIRE**. Our analysis shows that query-language preference is an independent behavioral axis shaped by the interaction between semantic alignment, lexical anchoring, and the language composition of retrieval supervision.

Table 2: Retrieval results across three datasets for all 31 retrievers. MLQA (7 languages) and XQuAD (12 languages) are evaluated at k=20; Belebele (122 languages) is evaluated at k=200. All values are percentages (raw scores $\times$100). **Base** is standard nDCG; **Lang** is language-aware nDCG (Lang-nDCG). Per column, the best value is **bold** and the second best is underlined.

Model	MLQA (7)			XQuAD (12)			Belebele (122)		
	nDCG@20		LPR	nDCG@20		LPR	nDCG@200		LPR
	Base	Lang		Base	Lang		Base	Lang	
Dense Retrievers									
Qwen3-Embedding-8B	**68.64**	64.97	53.00	92.38	91.06	76.46	**74.35**	**74.41**	73.22
Qwen3-Embedding-4B	66.05	62.64	52.30	91.57	90.04	73.92	62.58	62.75	65.09
Qwen3-Embedding-0.6B	55.16	53.79	53.65	79.44	79.55	72.78	43.09	43.61	53.82
llama-embed-nemotron-8b	68.31	**66.82**	64.55	94.37	93.54	83.91	64.68	65.12	71.10
llama-nemotron-embed-1b-v2	63.61	63.23	63.99	94.56	**93.86**	84.11	38.98	40.36	72.15
pplx-embed-v1-4b	67.07	62.74	46.65	**94.94**	90.24	51.51	58.12	57.05	25.59
pplx-embed-v1-0.6b	61.54	60.11	58.06	89.57	88.17	72.11	49.31	49.24	43.09
jina-embeddings-v5-small	62.07	58.66	49.55	90.67	88.19	65.47	58.85	58.63	51.63
jina-embeddings-v5-nano	61.25	57.79	48.42	90.13	87.90	67.18	55.81	55.73	53.26
jina-embeddings-v3	28.72	28.24	42.00	68.96	65.69	36.65	29.20	29.04	24.98
gte-Qwen2-7B-instruct	60.34	58.52	57.98	88.93	85.86	59.16	58.20	57.31	34.72
gte-Qwen2-1.5B-instruct	51.27	55.24	80.78	76.66	79.72	93.08	38.67	39.74	66.11
gte-multilingual-base	54.12	52.65	51.82	83.31	80.50	53.77	62.37	60.71	18.04
snowflake-arctic-embed-l-v2.0	54.97	55.65	63.79	85.31	85.28	78.62	49.71	50.20	58.52
snowflake-arctic-embed-m-v2.0	45.14	48.87	71.93	65.56	69.93	86.41	29.48	30.96	68.23
multilingual-e5-large	42.63	52.49	**96.15**	69.92	75.55	**99.92**	42.40	43.55	62.76
multilingual-e5-base	40.23	49.86	94.73	61.68	68.73	99.68	35.68	37.01	62.10
multilingual-e5-small	35.97	46.42	95.91	51.11	60.29	99.73	23.66	24.76	44.14
harrier-oss-v1-0.6b	43.77	53.15	95.66	62.60	69.64	99.75	33.00	35.19	81.70
harrier-oss-v1-270m	42.48	51.08	94.17	60.61	67.79	99.56	32.55	34.71	80.99
voyage-4-nano	62.54	59.66	52.86	91.89	88.46	57.84	63.43	62.59	37.41
embeddinggemma-300m	61.22	60.80	67.44	86.85	87.33	86.35	49.06	49.28	53.22
bge-m3	56.65	55.43	53.79	88.06	85.94	64.42	59.41	58.81	39.25
nomic-embed-text-v2-moe	52.89	55.03	67.59	82.36	83.37	82.01	45.12	46.08	64.86
granite-embedding-278m	43.56	44.65	58.36	70.38	70.83	64.49	35.14	35.52	44.14
paraphrase-multilingual-MiniLM	34.92	32.65	36.88	70.55	64.98	23.47	24.13	23.63	11.63
LaBSE	28.41	26.85	37.53	64.70	59.99	24.03	42.18	40.88	8.96
Sparse Retrievers									
opensearch-neural-sparse-v1	29.98	38.21	91.03	29.19	39.88	95.24	15.90	17.78	71.66
BM25 (XLM-R tokenizer)	27.92	38.23	93.68	23.31	37.77	98.75	7.27	10.27	**89.76**
Late-Interaction Retrievers									
jina-colbert-v2	62.09	61.78	60.42	89.42	88.75	79.10	44.70	45.66	66.93
LFM2-ColBERT-350M	39.48	49.00	90.73	32.74	44.02	98.41	10.06	12.44	87.75

5.1 Main Results

nDCG–LPR Mismatch Table 2 reports the main results for all 31 retrievers. The central observation is that standard language-agnostic retrieval quality does not reliably predict whether a retriever prioritizes query-language passages. Across the three datasets, the association between standard nDCG and LPR is weakly negative: the Pearson/Spearman correlations are -0.28/-0.30 on MLQA, -0.38/-0.47 on XQuAD, and -0.29/-0.28 on Belebele. This shows that semantic retrieval quality and query-language preference form distinct behavioral axes. The Lang-nDCG results further reflect this tension. Models with similar Base nDCG can receive different language-aware scores depending on whether they prioritize query-language passages. For example, on MLQA, Qwen3-Embedding-8B achieves higher Base nDCG than llama-embed-nemotron-8b, but lower Lang-nDCG, consistent with its much weaker LPR. Thus, high semantic retrieval quality can coexist with weak query-language preference, where relevant content is retrieved but not necessarily in the query language.

Paradigm-Level Patterns The nDCG–LPR mismatch appears differently across retrieval paradigms. Dense retrievers show large variation across model families: recent large-scale embedding models such as Qwen3-Embedding and llama-nemotron achieve strong nDCG, whereas the

multilingual-e5 family shows much stronger LPR despite lower nDCG. Sparse retrievers show a more consistent pattern. BM25 and OpenSearch-NSE achieve high LPR, but their standard nDCG is much lower than that of the strongest dense and late-interaction retrievers. This is an expected behavior because sparse retrieval relies heavily on lexical overlap, which naturally favors query-language evidence but limits cross-lingual semantic matching. Late-interaction retrievers reveal another trade-off. jina-colbert-v2 achieves competitive semantic retrieval, suggesting that token-level matching can provide fine-grained multilingual relevance signals. However, LFM2-ColBERT preserves query language more strongly while showing weaker semantic retrieval.

Role of Training Dataset Training Dataset offers a plausible explanation for these patterns. Qwen3-Embedding and llama-nemotron models construct multilingual fine-tuning data with cross-lingual relevance pairs, where queries and positive passages are written in different languages [40, 50]. Such supervision encourages semantically equivalent passages across languages to be closely aligned, which improves cross-lingual retrieval but can make translated relevant passages overly competitive with the query-language passage. By contrast, multilingual-e5 follows multi-monolingual contrastive training, where query–positive pairs are constructed within the same language [32]. This structure can preserve language identity more strongly, consistent with the high LPR of the family. A similar pattern appears in late-interaction retrieval: jina-colbert-v2 uses cross-lingual relevance supervision during retrieval fine-tuning [46], whereas LFM2-ColBERT is initialized from a multilingual model but fine-tuned on English data [47]. This suggests that the language composition of training data plays a central role in shaping the trade-off between semantic alignment and language preservation.

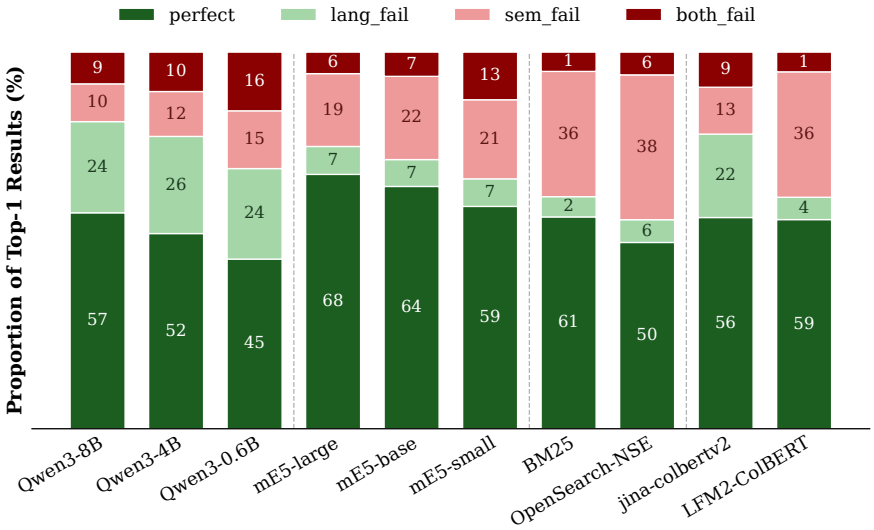

Figure 3: Failure decomposition across representative retrievers. Each bar partitions the top-ranked result into `perfect`, `lang_fail`, `sem_fail`, and `both_fail`.

5.2 Analysis of Top-1 Ranked Results

Decomposition Setup Figure 3 decomposes each retriever's top-ranked result into the four outcomes defined in Section 3.3. The reported proportions are macro-averaged over the three MLAIRE datasets - Belebele, XQuAD, and MLQA. We include representative models from the main behavioral regimes in Table 2: Qwen3-Embedding as a semantically strong dense retriever, multilingual-e5 (mE5) as a language-preserving dense retriever, BM25 and OpenSearch-NSE as sparse retrievers, and jina-colbert-v2 and LFM2-ColBERT as late-interaction retrievers.

Semantic Match with Query-Language Mismatch Qwen3-Embedding-8B and jina-colbert-v2 frequently retrieve semantically relevant passages at the top rank, as shown by their high combined `perfect` and `lang_fail` rates. However, a substantial share of these semantic matches are `lang_fail`: the model retrieves the correct content, but in a language different from the query. Standard nDCG counts this outcome as successful because the retrieved content is semantically relevant. This explains why strong semantic retrieval does not imply strong query-language preference.

Query-Language Retrieval with Semantic Failures The opposite pattern appears for mE5 and sparse retrievers. These models more often preserve the query language, but their non-perfect outcomes are more frequently semantic failures. In particular, BM25 and OpenSearch-NSE rarely produce `lang_fail`, yet they show much larger `sem_fail` and `both_fail` proportions. This shows that strong query-language preference does not necessarily imply successful semantic retrieval. For sparse retrievers, high language preservation can partly reflect lexical anchoring, which favors same-language passages even when they do not contain the correct content.

Table 3: Per-query-language LPR on XQuAD. Resource tiers follow Table 5

Model	High-resource									Mid-resource			AVG
	de	es	en	vi	hi	ar	zh	tr	ru	ro	th	el	
Qwen3-Embedding-8B	63.61	60.67	53.28	68.24	73.11	71.01	61.93	90.67	93.70	97.98	86.30	96.97	76.46
Qwen3-Embedding-4B	56.05	56.05	54.87	61.43	66.30	70.34	67.65	84.87	91.26	87.23	94.12	97.23	73.95
Qwen3-Embedding-0.6B	54.96	59.33	51.34	68.99	72.35	75.80	74.87	73.36	80.67	81.60	93.19	86.55	72.75
multilingual-e5-large	**99.92**	**100.00**	**99.75**	**99.92**	**100.00**	**99.92**	**100.00**	**100.00**	99.83	**100.00**	99.92	**99.75**	**99.92**
multilingual-e5-base	99.66	99.83	**99.75**	99.50	99.50	99.66	**100.00**	99.66	99.75	99.92	99.33	99.58	99.68
multilingual-e5-small	99.83	99.83	99.33	**99.92**	99.83	99.75	**100.00**	99.33	**99.92**	99.33	**100.00**	99.66	99.73
BM25	97.56	97.98	97.82	99.58	**100.00**	99.66	99.33	97.06	98.82	98.32	99.58	98.66	98.70
OpenSearch-NSE	89.58	92.27	91.09	97.73	99.24	98.99	98.99	95.63	98.66	93.03	80.67	97.73	94.47
Jina-ColBERT-v2	63.28	63.61	89.58	82.86	79.24	79.66	92.61	75.63	81.93	66.81	88.15	75.04	78.20
LFM2-ColBERT-350M	96.47	97.98	91.43	99.58	99.33	99.16	99.83	99.16	99.41	98.82	95.97	96.72	97.82

5.3 Query-Language Variation in LPR

Variation Across Query Languages Aggregated LPR can hide substantial variation across query languages, so we report per-query-language LPR on XQuAD in Table 3. The XQuAD languages in the table include high-resource languages from de to ru and mid-resource languages from ro to el, following the resource tiers in Table 5. Among semantically strong retrievers, the Qwen3-Embedding family and jina-colbert-v2 show the clearest language-dependent variation. For Qwen3-Embedding, LPR is particularly low for several high-resource languages such as German (de), Spanish (es), English (en), Vietnamese (vi), and Chinese (zh), while it is much higher for mid-resource languages. jina-colbert-v2 also varies substantially across query languages, with low LPR for German (de), Spanish (es), Romanian (ro), and Greek (el), but much higher LPR for English (en), Chinese (zh), and Thai (th). These results show that query-language preference is not uniform across languages, even within the same model family.

Stable LPR Across Languages Other retrievers show more stable LPR patterns. The multilingual-e5 family, BM25, and LFM2-ColBERT remain close to the LPR ceiling for most XQuAD query languages, while OpenSearch-NSE is generally high but less uniform. However, high LPR does not by itself indicate strong semantic retrieval, since it can also arise from lexical anchoring or limited cross-lingual matching. Thus, per-language LPR identifies which query languages are vulnerable, but not where the model retrieves from when query-language preference fails.

5.4 Directional Query-Language Mismatch by Language Group

To analyze where query-language preference fails, we group the 122 Belebele languages into ten macro language groups following the classification in Appendix E. Figure 4 reports the transition distribution over LPR-failure cases. Columns denote the query-language group, rows denote the retrieved document-language group, and each cell shows the proportion of non-query-language selections assigned to the corresponding document group.

Diagonal Concentration Across Language Groups Non-query-language retrieval is highly structured across macro language groups. The dominant pattern is diagonal concentration: when retrievers fail to select the query language passage, they often select a semantically equivalent passage from another language in the same macro group. For example, Germanic, Slavic-Baltic, Arabic-Semitic, South Asian, Southeast Asian, and Sub-Saharan African queries frequently retrieve documents from their own macro groups, and this trend is visible across all retrievers. This indicates that query-language preference failures are not random language confusions, but are structured by language-family, script, or regional affinity.

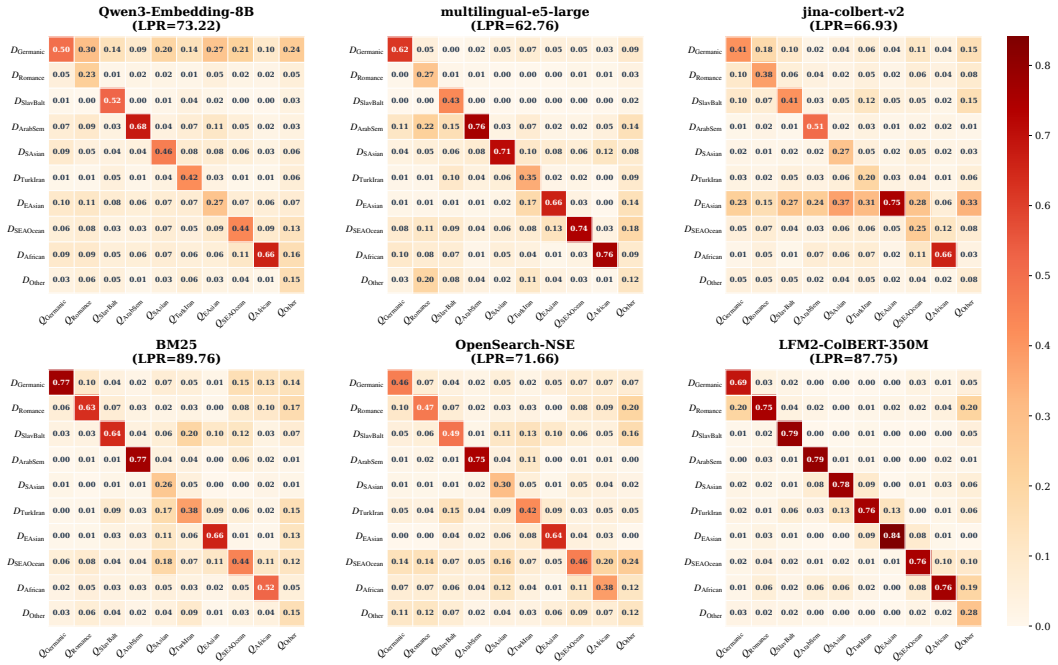

Figure 4: Transition distributions of LPR-failure cases across Belebele macro language groups. Columns denote the query-language group and rows denote the retrieved document-language group. Each cell reports the proportion of wrong-language selections assigned to the corresponding document group.

Model-Specific Directional Patterns The strength and direction of this structure differ across retrievers. Qwen3-Embedding-8B and multilingual-e5-large show clear within-group concentration for many query groups, but also exhibit cross-group attraction, such as Romance queries frequently retrieving Germanic documents. BM25, OpenSearch-NSE, and LFM2-ColBERT show the strongest diagonal concentration among the displayed models, consistent with their high LPR and stronger tendency to preserve language or script-level similarity. By contrast, Jina-ColBERT-v2 exhibits a more asymmetric pattern. Its failures are often attracted toward East Asian documents across many query groups, rather than being explained solely by same-group substitution. These differences suggest that non-query-language retrieval is shaped not only by architecture, but also by the language composition and supervision signals used during retrieval training.

Overall, the macro-group transition analysis complements per-language LPR. Per-language LPR identifies which query languages are vulnerable, while Figure 4 shows which document-language groups are selected when query-language preference fails. The results show that LPR failures are often structured by group-level linguistic affinity and model-specific target-language attraction, rather than by random selection among languages.

6 Conclusion

We introduce **MLAIRE**, a multilingual language-aware retrieval evaluation protocol that disentangles semantic retrieval quality from query-language preference. Across 31 retrievers, we show that standard retrieval metrics do not reliably indicate whether a model retrieves evidence in the query language. Using MLAIRE, dense, sparse, and late-interaction retrievers exhibit distinct trade-offs: semantically strong models can retrieve relevant evidence in a non-query language, while language-preserving models may lack robust cross-lingual semantic coverage. Through top-1 failure decomposition, per-language LPR analysis, and macro-group transition analysis, we further show that non-query-language retrieval is often structured. These findings highlight the need to evaluate multilingual retrievers not only by what they retrieve, but also by whether their retrieval behavior aligns with users' language expectations.

References

[1] Jian-Yun Nie and Fuman Jin. A multilingual approach to multilingual information retrieval. In *Workshop of the Cross-Language Evaluation Forum for European Languages*, pages 101–110. Springer, 2002.

[2] Carol Peters, Martin Braschler, and Paul Clough. *Multilingual information retrieval: From research to practice*. Springer, 2012.

[3] Eugene Yang, Dawn Lawrie, and James Mayfield. Distillation for multilingual information retrieval. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR 2024, page 2368–2373. ACM, 2024. doi: 10.1145/3626772.3657955. URL `http://dx.doi.org/10.1145/3626772.3657955`.

[4] Eugene Yang, Thomas Jänich, James Mayfield, and Dawn Lawrie. Language fairness in multilingual information retrieval. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2487–2491, 2024.

[5] Jinrui Yang, Fan Jiang, and Timothy Baldwin. Language bias in information retrieval: The nature of the beast and mitigation methods. *arXiv preprint arXiv:2509.06195*, 2025.

[6] Jeonghyun Park and Hwanhee Lee. Investigating language preference of multilingual rag systems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5647–5675, 2025.

[7] Wei Liu, Sony Trenous, Leonardo F. R. Ribeiro, Bill Byrne, and Felix Hieber. Xrag: Cross-lingual retrieval-augmented generation, 2025. URL `https://arxiv.org/abs/2505.10089`.

[8] Dayeon Ki, Marine Carpuat, Paul McNamee, Daniel Khashabi, Eugene Yang, Dawn Lawrie, and Kevin Duh. Linguistic nepotism: Trading-off quality for language preference in multilingual rag. *arXiv preprint arXiv:2509.13930*, 2025.

[9] Uma Roy, Noah Constant, Rami Al-Rfou, Aditya Barua, Aaron Phillips, and Yinfei Yang. Lareqa: Language-agnostic answer retrieval from a multilingual pool. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 5919–5930, 2020.

[10] Martin Braschler. Clef 2002—overview of results. In *Workshop of the Cross-Language Evaluation Forum for European Languages*, pages 9–27. Springer, 2002.

[11] Dawn Lawrie, James Mayfield, Eugene Yang, Andrew Yates, Sean MacAvaney, Ronak Pradeep, Scott Miller, Paul McNamee, and Luca Soldani. Neuclirbench: A modern evaluation collection for monolingual, cross-language, and multilingual information retrieval. *arXiv preprint arXiv:2511.14758*, 2025.

[12] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogenous benchmark for zero-shot evaluation of information retrieval models, 2021. URL `https://arxiv.org/abs/2104.08663`.

[13] Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. Mteb: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, 2023.

[14] Ivan Vulić and Marie-Francine Moens. Monolingual and cross-lingual information retrieval models based on (bilingual) word embeddings. In *Proceedings of the 38th international ACM SIGIR conference on research and development in information retrieval*, pages 363–372, 2015.

[15] Seungyoon Lee, Minhyuk Kim, Seongtae Hong, Youngjoon Jang, Dongsuk Oh, and Heuiseok Lim. Clear: Cross-lingual enhancement in alignment via reverse-training. *arXiv preprint arXiv:2604.05821*, 2026.

[16] Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, et al. Mmteb: Massive multilingual text embedding benchmark. *arXiv preprint arXiv:2502.13595*, 2025.

[17] Seongtae Hong, Youngjoon Jang, Jungseob Lee, Hyeonseok Moon, and Heuiseok Lim. Improving semantic proximity in information retrieval through cross-lingual alignment. *arXiv preprint arXiv:2604.05684*, 2026.

[18] Zewen Chi, Li Dong, Furu Wei, Nan Yang, Saksham Singhal, Wenhui Wang, Xia Song, Xian-Ling Mao, He-Yan Huang, and Ming Zhou. Infoxlm: An information-theoretic framework for cross-lingual language model pre-training. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3576–3588, 2021.

[19] Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic bert sentence embedding. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 878–891, 2022.

[20] Michael Buckland and Fredric Gey. The relationship between recall and precision. *Journal of the American society for information science*, 45(1):12–19, 1994.

[21] Hamed Valizadegan, Rong Jin, Ruofei Zhang, and Jianchang Mao. Learning to rank by optimizing ndcg measure. *Advances in neural information processing systems*, 22, 2009.

[22] CSA Research. Consumers prefer their own language, 2020. URL `https://csa-research.com/l/media/Consumers-Prefer-their-Own-Language`. Accessed: 2026-04-12.

[23] Punam Singh, Carly Wight, Olcan Sercinoglu, David Wilson, Artem Boytsov, and Manish Raizada. Language preferences on websites and in google searches for human health and food information. *Journal of medical Internet research*, 9(2):e625, 2007.

[24] Chenjun Ling, Ben Steichen, and Alexander G. Choulos. A comparative user study of interactive multilingual search interfaces. In *Proceedings of the 2018 Conference on Human Information Interaction & Retrieval*, CHIIR '18, page 211–220, New York, NY, USA, 2018. Association for Computing Machinery. ISBN 9781450349253. doi: 10.1145/3176349.3176383. URL `https://doi.org/10.1145/3176349.3176383`.

[25] Ben Steichen and Ryan Lowe. How do multilingual users search? an investigation of query and result list language choices. *Journal of the Association for Information Science and Technology*, 72(6):759–776, 2021. doi: https://doi.org/10.1002/asi.24443. URL `https://asistdl.onlinelibrary.wiley.com/doi/abs/10.1002/asi.24443`.

[26] Sunny Nahar, Ali Tawfiq, and Danny Sullivan. How google search handles multilingual searches, September 2023. URL `https://developers.google.com/search/blog/2023/09/multilingual-searches`.

[27] Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4623–4637, 2020.

[28] Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL `https://arxiv.org/abs/2412.15115`.

[29] Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabsa. The belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 749–775, 2024.

[30] Marta R Costa-Jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.

[31] Patrick Lewis, Barlas Oguz, Ruty Rinott, Sebastian Riedel, and Holger Schwenk. Mlqa: Evaluating cross-lingual extractive question answering. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 7315–7330, 2020.

[32] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Multilingual e5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*, 2024.

[33] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*, 4(5), 2024.

[34] Xin Zhang, Yanzhao Zhang, Dingkun Long, Wen Xie, Ziqi Dai, Jialong Tang, Huan Lin, Baosong Yang, Pengjun Xie, Fei Huang, et al. mgte: Generalized long-context text representation and reranking models for multilingual text retrieval. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1393–1412, 2024.

[35] Puxuan Yu, Luke Merrick, Gaurav Nuti, and Daniel Campos. Arctic-embed 2.0: Multilingual retrieval without compromise. *arXiv preprint arXiv:2412.04506*, 2024.

[36] Zach Nussbaum, John X Morris, Brandon Duderstadt, and Andriy Mulyar. Nomic embed: Training a reproducible long context text embedder. *arXiv preprint arXiv:2402.01613*, 2024.

[37] Henrique Schechter Vera, Sahil Dua, Biao Zhang, Daniel Salz, Ryan Mullins, Sindhu Raghuram Panyam, Sara Smoot, Iftekhar Naim, Joe Zou, Feiyang Chen, et al. Embeddinggemma: Powerful and lightweight text representations. *arXiv preprint arXiv:2509.20354*, 2025.

[38] Saba Sturua, Isabelle Mohr, Mohammad Kalim Akram, Michael Günther, Bo Wang, Markus Krimmel, Feng Wang, Georgios Mastrapas, Andreas Koukounas, Nan Wang, et al. jina-embeddings-v3: Multilingual embeddings with task lora. *arXiv preprint arXiv:2409.10173*, 2024.

[39] Mohammad Kalim Akram, Saba Sturua, Nastia Havriushenko, Quentin Herreros, Michael Günther, Maximilian Werk, and Han Xiao. jina-embeddings-v5-text: Task-targeted embedding distillation. *arXiv preprint arXiv:2602.15547*, 2026.

[40] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, et al. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025.

[41] Gabriel de Souza P Moreira, Radek Osmulski, Mengyao Xu, Ronay Ak, Benedikt Schifferer, and Even Oldridge. Nv-retriever: Improving text embedding models with effective hard-negative mining. *arXiv preprint arXiv:2407.15831*, 2024.

[42] Sedigheh Eslami, Maksim Gaiduk, Markus Krimmel, Louis Milliken, Bo Wang, and Denis Bykov. Diffusion-pretrained dense and contextual embeddings. *arXiv preprint arXiv:2602.11151*, 2026.

[43] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 8440–8451, 2020.

[44] Andrzej Białecki, Robert Muir, Grant Ingersoll, and Lucid Imagination. Apache Lucene 4. In *SIGIR 2012 Workshop on Open Source Information Retrieval*, page 17. sn, 2012.

[45] Zhichao Geng, Yiwen Wang, Dongyu Ru, and Yang Yang. Towards competitive search relevance for inference-free learned sparse retrievers, 2025. URL `https://arxiv.org/abs/2411.04403`.

[46] Rohan Jha, Bo Wang, Michael Günther, Georgios Mastrapas, Saba Sturua, Isabelle Mohr, Andreas Koukounas, Mohammad Kalim Wang, Nan Wang, and Han Xiao. Jina-ColBERT-v2: A general-purpose multilingual late interaction retriever. In Jonne Sälevä and Abraham Owodunni, editors, *Proceedings of the Fourth Workshop on Multilingual Representation Learning (MRL*

2024), pages 159–166, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.mrl-1.11. URL https://aclanthology.org/2024.mrl-1.11/.

[47] Alexander Amini, Anna Banaszak, Harold Benoit, Arthur Böök, Tarek Dakhran, Song Duong, Alfred Eng, Fernando Fernandes, Marc Härkönen, Anne Harrington, et al. Lfm2 technical report. *arXiv preprint arXiv:2511.23404*, 2025.

[48] Omar Khattab and Matei Zaharia. Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*, pages 39–48, 2020.

[49] Keshav Santhanam, Omar Khattab, Christopher Potts, and Matei Zaharia. Plaid: an efficient engine for late interaction retrieval. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pages 1747–1756, 2022.

[50] Yauhen Babakhin, Radek Osmulski, Ronay Ak, Gabriel Moreira, Mengyao Xu, Benedikt Schifferer, Bo Liu, and Even Oldridge. Llama-embed-nemotron-8b: A universal text embedding model for multilingual and cross-lingual tasks, 2025. URL https://arxiv.org/abs/2511.07025.

[51] David Ifeoluwa Adelani, Hannah Liu, Xiaoyu Shen, Nikita Vassilyev, Jesujoba Alabi, Yanke Mao, Haonan Gao, and En-Shiun Annie Lee. Sib-200: A simple, inclusive, and big evaluation dataset for topic classification in 200+ languages and dialects. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 226–245, 2024.

Appendices

A Limitations

Scope of query-language preference MLAIRE treats the query-language version of a semantically equivalent passage as the preferred retrieval target when such a passage is available. This is appropriate for many user-facing search and RAG scenarios, where users expect to read and verify evidence in the language of their query. However, LPR should not be interpreted as a universal utility measure. For bilingual users, code-switching contexts, or domains where high-quality information is more abundant in a non-query language, cross-lingual evidence may be equally useful or even preferable. Therefore, LPR and Lang-nDCG should be used as complementary diagnostics rather than replacements for standard semantic retrieval metrics.

Dataset scope Our evaluation is built from Belebele, XQuAD, and MLQA, which provide fully or partially parallel QA-style passages. This construction creates a controlled setting where semantic relevance can be separated from query-language preference. However, it may not fully represent open-domain mixed-language corpora, where documents can differ in length, style, topical coverage, translation quality, and native-language authorship. Future work should extend the protocol to ad hoc retrieval collections with native relevance judgments across languages and naturally occurring mixed-language document pools.

Partial parallelism for MLQA Belebele and XQuAD are fully parallel, whereas MLQA is only partially parallel. As a result, the number of semantically equivalent language alternatives differs across datasets. This can affect both LPR and Lang-nDCG because the set of competing translations is smaller in MLQA.

Considerations for new metrics LPR measures whether the query-language version of the target content receives the highest score among semantically equivalent alternatives. A model could increase LPR by over-weighting language-identification or surface-form cues after matching the content group, without improving semantic retrieval. This is why MLAIRE reports LPR together with standard nDCG, Lang-nDCG, and the top-1 4-way decomposition. Future diagnostic controls could include language-confusable distractors, script-normalized variants, and adversarial same-language passages that share surface cues but not content.

RAG motivation experiment The RAG experiment in Section 2 is intended as a motivating analysis. It controls semantic content by comparing English and query-language gold passages, but the current presentation does not exhaustively vary retrieval depth, passage chunking, prompt templates, decoding settings, or answer-normalization choices. These details can affect answer accuracy and language coherence. Future work should evaluate whether the same pattern holds under retrieved rather than gold passages, multiple generators, and broader decoding configurations.

Fairness and language bias Query-language preference overlaps with language fairness in retrieval, but the objectives are not identical. A model with high LPR may still underserve users if it retrieves same-language but semantically weak evidence. Conversely, a strong cross-lingual retriever may receive lower LPR even when it benefits bilingual users or users seeking content from other language communities. Accordingly, MLAIRE should be used to diagnose retrieval behavior, not to enforce monolingual retrieval in all multilingual settings.

B Ethical Considerations

Dataset licenses and reuse MLAIRE is constructed from public multilingual datasets. Belebele is released under CC-BY-SA 4.0, XQuAD is distributed with a CC-BY-SA 4.0 license file, and MLQA states that its dataset, derived from Wikipedia paragraphs, is licensed under CC-BY-SA 3.0. Users of MLAIRE should follow the attribution and share-alike requirements of the underlying datasets.

Representativeness Although the benchmark covers many languages, coverage is still uneven across language families, scripts, and resource levels. High-resource languages and translated benchmark content may be overrepresented relative to naturally occurring multilingual search scenarios. Results should therefore be interpreted as controlled measurements of retrieval behavior, not as comprehensive evidence of equitable performance across all linguistic communities.

Responsible use of LPR LPR should not be used as a standalone deployment objective. It is most appropriate when the application expects users to read or verify evidence in the same language as their query. It may be less appropriate for bilingual users, code-switching users, cross-lingual research workflows, or settings where the best available evidence is naturally written in another language. Operational use should combine LPR with semantic retrieval metrics and application-specific user needs.

Fairness trade-offs Optimizing for query-language preference can improve accessibility for users who need same-language evidence, but it can also penalize systems that intentionally retrieve cross-lingual evidence for broader coverage. Conversely, high LPR with low semantic retrieval quality can still harm users by returning readable but incorrect or irrelevant evidence. The intended use of MLAIRE is diagnostic: it helps separate semantic retrieval quality from language preference so that developers can understand trade-offs rather than optimize a single metric blindly.

Positive impact MLAIRE can help developers diagnose whether multilingual retrievers provide evidence in a language users can read and verify, improving accessibility and transparency in multilingual search and RAG systems.

Potential negative impact If LPR is optimized in isolation, systems may over-prioritize same-language passages even when cross-lingual evidence is more complete or reliable. We therefore recommend using LPR only together with semantic retrieval metrics and application-specific user needs.

C Instruction Templates

Table 4 lists the query and passage prefixes used for each dense retriever, grouped by model family. When a model exposes MTEB-style task prompts, we use those prompts; otherwise we fall back to the prefixes below, which match each model's original release recipe. For sparse and multi-vector retrievers, no instruction template is applied.

Table 4: Query and passage prefixes used for dense retrievers. \n denotes a newline separator

Model family	Query prefix	Passage prefix
multilingual-e5 {small, base, large}	`query:`	`passage:`
Nemotron-Embed-1B-v2	`query:`	`passage:`
Snowflake Arctic-L-v2.0	`query:`	–
Nomic-Embed-v2-MoE	`search_query:`	`search_document:`
Jina-v3	`Represent the query for retrieving evidence documents:`	`Represent the document for retrieval:`
Voyage-4-nano	`Represent the query for retrieving supporting documents:`	`Represent the document for retrieval:`
Jina-v5 {small, nano}	`Query:`	`Document:`
EmbeddingGemma-300M	`task: search result | query:`	`title: none | text:`
Harrier-OSS-v1 {270M, 0.6B}	`Instruct: Given a question, retrieve passages that answer the question\n Query:`	–
Qwen3-Embedding {0.6B, 4B, 8B}	`Instruct: Given a question, retrieve passages that answer the question\n Query:`	–
llama-embed-nemotron-8B	`Instruct: Given a question, retrieve passages that answer the question\n Query:`	–
llama-nemotron-embed-1b-v2	`query:`	`passage:`
gte-Qwen2-{1.5B, 7B}-instruct	`Instruct: Given a web search query, retrieve relevant passages that answer the query Query:`	–
bge-m3	–	–
gte-multilingual-base	–	–
pplx-embed-v1 {0.6B, 4B}	–	–
granite-embedding-278M-multilingual	–	–
LaBSE	–	–
paraphrase-mMiniLM-L12	–	–

D Infrastructure and Reproducibility

Hardware All experiments were run on a single workstation with four NVIDIA RTX A6000 (48 GB) GPUs and 1 TB of system RAM.

Software environments We use Python 3.10 and PyTorch 2.8 with CUDA 12.1. Sparse and multi-vector retrievers depend on distinct indexing toolchains (`bm25s`, `pylate`). To avoid dependency collisions while keeping the evaluator itself identical, we partition the retriever pool across three isolated virtual environments with pinned requirements; all three import the same evaluation core.

Runtime End-to-end evaluation time varies by retriever architecture and dataset size. Dense retrievers require corpus encoding followed by top-k retrieval, sparse retrievers require indexing with their respective sparse representations, and late-interaction models require PLAID indexing. In total, the reported experiments required approximately 50 GPU-hours and under 1 CPU-hour (BM25 baseline). The largest cost comes from encoding/indexing the Belebele 122-language corpus across all retrievers.

E Language Classification

Resource-level classification For analyses that require language resource levels, we primarily follow the resource classes provided with SIB-200 [51], which is based on FLORES-200 [30] and includes Joshi-style resource categories. We map classes 0–2 to low-resource, class 3 to mid-resource, and classes 4–5 to high-resource languages. For languages or script variants without a Joshi-style class, we use the NLLB-200 [30] high/low-resource distinction as a fallback, where low-resource languages are defined by having fewer than 1M publicly available deduplicated bitext sentence pairs. The resulting resource-level mapping is reported in Table 5.

Table 5: Language classification based on resource levels

Resource Level	FLORES-200 language codes
High	arb_Arab, cat_Latn, ces_Latn, deu_Latn, eng_Latn, eus_Latn, fin_Latn, fra_Latn, hin_Deva, hin_Latn, hrv_Latn, hun_Latn, ita_Latn, jpn_Jpan, kor_Hang, nld_Latn, pes_Arab, pol_Latn, por_Latn, rus_Cyrl, spa_Latn, srp_Cyrl, swe_Latn, tur_Latn, vie_Latn, zho_Hans
Mid	afr_Latn, arz_Arab, ben_Beng, ben_Latn, bul_Cyrl, ceb_Latn, dan_Latn, ell_Grek, est_Latn, heb_Hebr, ind_Latn, kat_Geor, kaz_Cyrl, lit_Latn, lvs_Latn, ron_Latn, slk_Latn, slv_Latn, tam_Taml, tgl_Latn, tha_Thai, ukr_Cyrl, urd_Arab, urd_Latn, uzn_Latn, zsm_Latn
Low	acm_Arab, als_Latn, amh_Ethi, apc_Arab, arb_Latn, ars_Arab, ary_arab, asm_Beng, azj_Latn, bam_Latn, bod_Tibt, ckb_Arab, fuv_Latn, gaz_Latn, grn_Latn, guj_Gujr, hat_Latn, hau_Latn, hye_Armn, ibo_Latn, ilo_Latn, isl_Latn, jav_Latn, kac_Latn, kan_Knda, kea_Latn, khk_Cyrl, khm_Khmr, kin_Latn, kir_Cyrl, lao_Laoo, lin_Latn, lug_Latn, luo_Latn, mal_Mlym, mar_Deva, mkd_Cyrl, mlt_Latn, mri_Latn, mya_Mymr, nob_Latn, npi_Deva, npi_Latn, nso_Latn, nya_Latn, ory_Orya, pan_Guru, pbt_Arab, plt_Latn, shn_Mymr, sin_Latn, sin_Sinh, sna_Latn, snd_Arab, som_Latn, sot_Latn, ssw_Latn, sun_Latn, swh_Latn, tel_Telu, tgk_Cyrl, tir_Ethi, tsn_Latn, tso_Latn, war_Latn, wol_Latn, xho_Latn, yor_Latn, zho_Hant, zul_Latn

Macro language groups For the directional analysis in Section 5.4, we group the 122 Belebele language variants into ten macro language groups. This grouping is not intended as a strict linguistic taxonomy. Rather, it is an analysis convenience that aggregates languages by broad family, script, and regional affinities so that group-level non-query-language retrieval patterns can be visualized. Table 6 lists the full mapping and the short labels used in Figure 4.

F Recall and Lang-Recall

Table 7 reports standard Recall and Lang-Recall at the same evaluation depths used in Table 2 (k=20 for MLQA and XQuAD, and k=200 for Belebele). Recall@k measures how many semantically relevant passages are retrieved within the top k, regardless of language. Lang-Recall@k restricts the relevant set to the query-language version of the target content and measures whether that version appears within the top k. Because the two metrics use different relevant sets, they capture different aspects of retrieval behavior. A retriever can achieve high standard Recall by retrieving many cross-lingual relevant passages while missing the query-language version, or achieve high Lang-Recall while retrieving only a small portion of the full multilingual relevant set.

High-LPR retrievers show higher Lang-Recall Retrievers with high LPR, including the multilingual-e5 and harrier-oss families, the sparse retrievers, and LFM2-ColBERT-350M, generally show higher Lang-Recall than standard Recall. For example, multilingual-e5-large obtains Lang-Recall@20 of 85.17 on MLQA and 99.87 on XQuAD, compared with standard Recall@20 of 43.97 and 72.59. BM25 shows an even larger gap on XQuAD, with Lang-Recall@20 of 98.56 compared with standard Recall@20 of 13.94. These results indicate that these retrievers often recover the query-language version of the target content, while retrieving fewer of the cross-lingual alternatives. This pattern is consistent with their high LPR, but reflects a different top-k perspective: Lang-Recall measures whether the query-language version is recovered within the retrieved set, whereas LPR measures which language version is preferred within the target content group.

Lower-LPR retrievers show smaller gaps Retrievers with lower or more moderate LPR, such as pplx-embed-v1-4b, gte-multilingual-base, jina-embeddings-v3, paraphrase-multilingual-MiniLM,

Table 6: Macro language groups used for the Belebele directional analysis. The short labels correspond to the group labels shown in Figure 4.

Short label	Macro group	FLORES-200 language codes
`Germanic`	Germanic / Anglophone-North European	afr_Latn, dan_Latn, deu_Latn, eng_Latn, isl_Latn, nld_Latn, nob_Latn, swe_Latn
`Romance`	Romance / Latin-European + Creole	cat_Latn, fra_Latn, ita_Latn, por_Latn, ron_Latn, spa_Latn, hat_Latn, kea_Latn
`SlavBalt`	Slavic-Baltic / Eastern European	bul_Cyrl, ces_Latn, hrv_Latn, lit_Latn, lvs_Latn, mkd_Cyrl, pol_Latn, rus_Cyrl, slk_Latn, slv_Latn, srp_Cyrl, ukr_Cyrl
`ArabSem`	Arabic-Semitic / MENA	acm_Arab, apc_Arab, arb_Arab, arb_Latn, ars_Arab, ary_arab, arz_Arab, heb_Hebr, mlt_Latn
`SAsian`	South Asian / Indo-Aryan-Dravidian	asm_Beng, ben_Beng, ben_Latn, guj_Gujr, hin_Deva, hin_Latn, kan_Knda, mal_Mlym, mar_Deva, npi_Deva, npi_Latn, ory_Orya, pan_Guru, sin_Latn, sin_Sinh, snd_Arab, tam_Taml, tel_Telu, urd_Arab, urd_Latn
`TurkIran`	Iranian-Turkic / Central-West Asian	azj_Latn, ckb_Arab, kaz_Cyrl, kir_Cyrl, pbt_Arab, pes_Arab, tgk_Cyrl, tur_Latn, uzn_Latn
`EAsian`	East Asian / CJK-Mongolic-Tibetic	bod_Tibt, zho_Hans, zho_Hant, jpn_Jpan, kor_Hang, khk_Cyrl
`SEAOcean`	Southeast Asian / Austronesian/Oceanic	ceb_Latn, ilo_Latn, ind_Latn, jav_Latn, kac_Latn, khm_Khmr, lao_Laoo, mri_Latn, mya_Mymr, plt_Latn, shn_Mymr, sun_Latn, tgl_Latn, tha_Thai, vie_Latn, war_Latn, zsm_Latn
`African`	Sub-Saharan African	amh_Ethi, bam_Latn, fuv_Latn, gaz_Latn, hau_Latn, ibo_Latn, kin_Latn, lin_Latn, lug_Latn, luo_Latn, nso_Latn, nya_Latn, sna_Latn, som_Latn, sot_Latn, ssw_Latn, swh_Latn, tir_Ethi, tsn_Latn, tso_Latn, wol_Latn, xho_Latn, yor_Latn, zul_Latn
`Other`	Other Europe / Caucasus / Americas	als_Latn, ell_Grek, est_Latn, eus_Latn, fin_Latn, grn_Latn, hun_Latn, hye_Armn, kat_Geor

and LaBSE, show smaller gaps between Recall and Lang-Recall. For instance, on XQuAD, pplx-embed-v1-4b obtains Recall@20 of 95.75 and Lang-Recall@20 of 98.88. On Belebele, the same model obtains Recall@200 of 56.67 and Lang-Recall@200 of 79.52. These results suggest that such retrievers distribute retrieval capacity more broadly across semantically equivalent passages in multiple languages, rather than concentrating as strongly on the query-language version. Accordingly, their standard Recall and Lang-Recall tend to move more closely together.

Largest gap in Belebele The difference between Recall and Lang-Recall is most pronounced on Belebele. Because each Belebele query can have up to 122 semantically relevant language variants, standard Recall has a much larger relevant set than in MLQA or XQuAD. A retriever that concentrates its top-k results on a subset of languages can therefore achieve high Lang-Recall while still obtaining relatively low standard Recall. For example, harrier-oss-v1-0.6b reaches Lang-Recall@200 of 96.24 on Belebele, while its standard Recall@200 is 30.73. This makes Belebele especially useful for separating query-language recovery from broader cross-lingual coverage.

Table 7: Recall, Lang-Recall, and LPR across the three datasets for all 31 retrievers. MLQA (7 languages) and XQuAD (12 languages) are evaluated at k=20; Belebele (122 languages) is evaluated at k=200. All values are percentages (raw scores $\times$100). **Recall** is standard Recall@k over the full multilingual relevant group. **Lang-Recall** restricts the relevant set to the query-language version of the target content and measures whether that single passage appears within the top-k. Per column, the best value is **bold** and the second best is underlined.

Model	MLQA (7)			XQuAD (12)			Belebele (122)		
	Recall@20		LPR	Recall@20		LPR	Recall@200		LPR
	Base	Lang		Base	Lang		Base	Lang	
Dense Retrievers									
Qwen3-Embedding-8B	**76.33**	80.87	53.00	93.84	99.20	76.46	**73.76**	96.22	73.22
Qwen3-Embedding-4B	73.76	78.48	52.30	92.99	98.78	73.92	61.60	92.81	65.09
Qwen3-Embedding-0.6B	62.28	70.57	53.65	80.46	96.33	72.78	41.02	85.38	53.82
llama-embed-nemotron-8b	75.71	83.34	64.55	95.62	99.66	83.91	63.34	**96.54**	71.10
llama-nemotron-embed-1b-v2	70.67	80.84	63.99	95.53	99.75	84.11	35.75	93.55	72.15
pplx-embed-v1-4b	74.43	78.24	46.65	**95.75**	98.88	51.51	56.67	79.52	25.59
pplx-embed-v1-0.6b	68.57	76.76	58.06	90.54	98.47	72.11	47.21	82.27	43.09
jina-embeddings-v5-small	69.96	74.67	49.55	92.06	97.73	65.47	57.77	89.40	51.63
jina-embeddings-v5-nano	69.14	73.59	48.42	91.53	97.69	67.18	54.69	88.00	53.26
jina-embeddings-v3	33.96	42.56	42.00	69.82	84.86	36.65	26.33	59.02	24.98
gte-Qwen2-7B-instruct	68.04	75.43	57.98	90.14	96.55	59.16	56.98	82.72	34.72
gte-Qwen2-1.5B-instruct	56.81	79.26	80.78	78.12	98.82	93.08	36.21	87.67	66.11
gte-multilingual-base	61.63	70.02	51.82	84.38	94.79	53.77	62.24	77.79	18.04
snowflake-arctic-embed-l-v2.0	60.97	74.55	63.79	86.63	97.76	78.62	47.14	90.60	58.52
snowflake-arctic-embed-m-v2.0	48.60	72.63	71.93	64.73	97.21	86.41	26.51	88.63	68.23
multilingual-e5-large	43.97	85.17	**96.15**	72.59	**99.87**	**99.92**	40.21	88.56	62.76
multilingual-e5-base	40.96	82.72	94.73	62.05	99.78	99.68	33.44	86.43	62.10
multilingual-e5-small	35.38	80.95	95.91	49.18	99.61	99.73	21.43	67.73	44.14
harrier-oss-v1-0.6b	45.67	**85.24**	95.66	63.76	99.82	99.75	30.73	96.24	81.70
harrier-oss-v1-270m	44.99	82.70	94.17	61.10	99.67	99.56	30.29	96.23	80.99
voyage-4-nano	70.57	76.59	52.86	93.05	98.16	57.84	62.20	90.91	37.41
embeddinggemma-300m	68.31	78.53	67.44	88.46	98.63	86.35	47.21	86.25	53.22
bge-m3	63.29	72.50	53.79	89.18	97.32	64.42	57.84	88.21	39.25
nomic-embed-text-v2-moe	58.40	76.15	67.59	83.26	98.07	82.01	42.27	92.11	64.86
granite-embedding-278m	49.05	64.35	58.36	71.24	92.88	64.49	32.96	79.16	44.14
paraphrase-multilingual-MiniLM	41.60	46.72	36.88	71.94	80.71	23.47	21.86	47.91	11.63
LaBSE	33.38	39.99	37.53	64.17	77.79	24.03	40.86	60.89	8.96
Sparse Retrievers									
opensearch-neural-sparse-v1	29.48	69.51	91.03	22.47	90.65	95.24	13.71	83.98	71.66
BM25 (XLM-R tokenizer)	25.73	72.98	93.68	13.94	98.56	98.75	5.22	92.82	**89.76**
Late-Interaction Retrievers									
jina-colbert-v2	66.04	77.31	60.42	89.86	97.42	79.10	41.80	91.93	66.93
LFM2-ColBERT-350M	37.31	80.29	90.73	25.49	93.89	98.41	7.88	85.10	87.75

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    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
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14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: We answered [N/A] because this research relies entirely on automated evaluations using existing benchmark datasets (Belebele, XQuAD, MLQA) and does not involve any crowdsourcing or human subject experiments.

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Answer: [Yes]

Justification: We use Qwen2.5-7B-Instruct only as the generator in the controlled RAG motivation experiment described in Section 2.2. LLMs are not used to construct MLAIRE, define the evaluation labels, or implement the proposed retrieval metrics. Any LLM assistance for writing or formatting does not affect the core methodology.

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